

Linear Regression

Learn how regression predicts a continuous value with a fitted weighted sum.



WHY THIS MATTERS

Start from what you already know

Regression versus classification, vectors, averages, loss, training, and testing

- 01 Explain continuous targets in plain language.
- 02 Use linear equation in a small worked example.
- 03 Complete the guided lab and verify regularization.

CORE CONCEPT

The vocabulary of this topic

Learn how regression predicts a continuous value with a fitted weighted sum.

continuous targets: A continuous target can take numerical values across an interval, such as price, temperature, or body-fat percentage.

linear equation: A linear prediction adds an intercept to feature values multiplied by learned coefficients.

residuals and RMSE: A residual is actual minus predicted; RMSE summarizes squared residuals in the target's original units.

regularization: Regularization penalizes complexity to improve stability and reduce sensitivity to noise.

FOUR CONNECTED IDEAS

How the concept works

01

**continuous
targets**



02

linear equation



03

**residuals and
RMSE**



04

regularization

$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$; training chooses coefficients that minimize squared residuals.

CORE CONCEPT

Read every part of the mechanism

$\hat{y} = \beta_0 + \beta_1 x_1 + \dots + \beta_n x_n$; training chooses coefficients that minimize squared residuals.

$\hat{y}$: Predicted continuous value.

β_0 : Intercept.

β_j : Coefficient for feature j .

x_j : Feature value j .

WORKED EXAMPLE

Worked example by hand

For hours=3, $\beta_0=40$ and $\beta_1=10$, predict 70; with actual 76, residual=6 and squared error=36.

01

Known values

02

Apply the mechanism

03

Interpret the result

RUN IT AND INSPECT THE RESULT

Map the concept to Python

```
intercept, coefficient, hours = 40, 10, 3  
prediction = intercept + coefficient * hours  
residual = 76 - prediction  
print(prediction, residual, residual**2)
```

OUTPUT

70 6 36

Each line corresponds to a concept already explained.

QUALITY GATE

Build the complete mental model



Ordinary least squares chooses coefficients that minimize the sum of squared residuals.



Closed-form solvers and gradient descent optimize the same objective through different procedures.



Check linearity, residual independence, roughly constant variance, and influential outliers.



Compare with a mean baseline; Ridge shrinks coefficients and Lasso can set some to zero.

FOUR CONNECTED IDEAS

Notebook readiness map

01

**continuous
targets**



02

linear equation



03

**residuals and
RMSE**



04

regularization

Use the concepts from this lesson to read and complete
<https://github.com/curiously/Al-Bootcamp/blob/master/03.model-development.ipynb>.

RISK REVIEW

Common beginner mistakes

- 01** **Calling RMSE an accuracy percentage even though it is an error measured in target units.**

- 02** **Interpreting a coefficient causally or without considering feature scaling.**

GUIDED LAB

Fit a baseline and linear regression, interpret coefficients, calculate RMSE, and inspect residuals.

01

02

03

04

Predict

Run

Inspect

Explain

SESSION SYNTHESIS

Keep the system, not isolated definitions.

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- 02 Use linear equation in a small worked example.
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NEXT EVIDENCE

A fitted regression pipeline with equation explanation, baseline comparison, diagnostics, and saved artifact.